Australian National University

# LoRA

Efficiently adapt a trained language model to a new domain or style without retraining the entire model from scratch.

## You will need

- a completed bigram model from an earlier lesson
- pen, pencil and grid paper
- some new text in a different style or domain

## Your goal

To create a lightweight “adaptation layer” that modifies your existing model’s behaviour for a new domain. **Stretch goal**: combine the base model and LoRA layer with different mixing ratios.

## Key idea

Low-Rank Adaptation (LoRA) allows you to specialise a language model by adding small adjustments rather than retraining everything. The LoRA layer is typically much smaller than the base model because you only track the *changes* from the base model, not the full weights.



## Algorithm

1. **choose an existing bigram model as the "base model"**
2. **train a LoRA layer**:
   - start with a new grid (same columns as the base model)
   - process your new domain-specific text using the same algorithm as *Basic Training*, but only include rows for words that appear in your new text
3. **apply the adaptation**:
   - as per *Basic Generation*, but add the counts from both grids (if current word is in the LoRA grid)
   - optionally scale the LoRA values up or down to control adaptation strength

## Example

**Base model** trained on general text:

| | saw | they | we | the | a | red |
|---|---|---|---|---|---|---|
| saw | | 2 | | 4 | 2 | 1 |
| they | 1 | | | 2 | 1 | |
| we | | | | 3 | | |
| the | | | 1 | | | |
| a | | | | | | 2 |
| red | | 1 | | | | |

**LoRA layer** trained on "I saw a red cat. I saw the red dog." (smaller — only 1 row):

| | saw | they | we | the | a | red |
|---|---|---|---|---|---|---|
| saw | | | | 1 | 1 | 2 |

**Combined model** (add counts):

| | saw | they | we | the | a | red |
|---|---|---|---|---|---|---|
| saw | | 2 | | 5 | 3 | 3 |
| they | 1 | | | 2 | 1 | |
| we | | | | 3 | | |
| the | | | 1 | | | |
| a | | | | | | 2 |
| red | | 1 | | | | |

- saw row:
  - [–,2,–,4,2,1] (base)
  - [–,–,–,1,1,2] (LoRA)
  - [–,2,–,5,3,3] (base + LoRA)
- red now equally likely as the after saw
- other rows: base + zero = unchanged
- LoRA is smaller: only 1 row vs 6 in base model